AI CANCER DETECTION SYSTEM — END-TO-END CLINICAL ARCHITECTURE

Hospital-Grade Multi-Organ AI Triage, Explainable XAI (Grad-CAM), & Multi-Agent Tumor Board

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1. CLINICAL CLIENT GATEWAYS

Role-Based Access & Ingestion

Physician & Staff Workstation

- Doctor / Radiologist Login
- PACS DICOM Multi-Slice Viewer
- Clinical Diagnostic Intake
- Interactive XAI Inspection

Patient Health Portal

- Patient Self-Service Access
- Plain-Language Diagnosis
- Stage Nutrition & Remedies
- Interactive Care Team Chat

Global Accessibility & UI

- DOM TreeWalker i18n Engine
- 11 Languages (En, Hi, Te, Ta...)
- Glassmorphic Responsive UI
- HIPAA Consent & Safety Alerts

2. CORE APPLICATION ENGINE

Flask 3.0 & Clinical Blueprints

Flask Web Service & REST API

- app.py Routing & Middlewares
- RBAC Authentication Sessions
- Secure DICOM/Image Ingest
- Synchronous Triage Response

Clinical Intelligence Suite

- hospital_core.py EMR Tracking
- health_suite.py Care Pathways
- health_guidance.py Staging
- Stage 0 to Stage IV Regimens

Integration & Reporting

- pdf_generator.py ReportLab
- email_service.py Patient Alerts
- google_drive_service.py Backup
- HIPAA Immutable Audit Trail

3. DEEP LEARNING & XAI

TensorFlow / Keras 2.18 Ensembles

EfficientNet-B0 Ensembles

- 224x224x3 Input Normalization
- Global Average Pooling 2D
- Dense(256) + Dropout(0.4)
- Softmax Multi-Class Scoring

Multi-Organ Detection Models

- Brain MRI (Glioma/Meningioma)
- Breast US (Benign/Malignant)
- Lung CT/X-Ray (Pulmonary)
- Skin Dermoscopy (Melanoma)

Explainability (Grad-CAM)

- Gradient-Class Activation Mapping
- Convolutional Feature Extraction
- Heatmap Overlay & Thresholding
- Visual Sanity & Trust Checks

4. MULTI-AGENT TUMOR BOARD

LLM Consensus & Diagnostics

Diagnostic Radiologist

- Radiological Density Scan
- Margin Definition & Spiculation
- Anatomical Slice Landmarks
- Calcification Analysis

Medical Oncologist

- TNM Staging Assessment
- Systemic Regimen Suggestions
- Biomarker & Genomic Advice
- Surveillance Cadence

Surgical Oncologist

- Resectability Assessment
- Operative Clearance & Margins
- Biopsy & Lymph Node Protocol
- Digital Report Certification